ADVANCED POWER THEFT DETECTION SYSTEM

Project Work done in partial fulfilment of the requirement for the degree of
Bachelor of Engineering in Electronics and Communication Engineering

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ABSTRACT

The Advanced Power Theft Detection System (APTD) presents an innovative solution for combating unauthorized power consumption within electrical systems.

In operation, the Advanced Power Theft Detection System utilizes two ESP8266 modules, designated as the primary and secondary nodes. The primary node continuously monitors current flow within the electrical circuit using a current sensor, while the secondary node serves as a baseline comparison

The parameter like voltage and current of the Advanced Power Theft Detection are done and monitor the energy consumption, thereby mitigating financial losses, enhancing system reliability, and promoting equitable access to electricity resources.

OVERVIEW

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INTRODUCTION

- Electricity theft is a very common problem, especially in our country.
- As our population is high so the use of electricity is tremendously high.
- There are many operational losses involve in the generation, transmission, and distribution of electrical energy.

OBJECTIVES

- **Detect Electricity Theft**

  Develop a system capable of detecting instances of electricity theft

- **Rigorous Testing and Evaluation**

  Conduct thorough testing, including simulations and practical implementation

- **Real-Time Monitoring**

  Implement continuous monitoring of power consumption in real-time.

- **Contribution to Efficiency and Reliability**

  Contribute to improving the overall efficiency and reliability of the electrical distribution network

LITERATURE SURVEY

TITLE OF THE PAPER WITH AUTHOR NAME	JOURNAL NAME/YEAR OF PUBLICATION	FINDINGS	LIMITATIONS
Title: A Literature Survey on Techniques of Power Theft Detection Authors: Prof. Vaibhav Baburao Magdum and Prof. Ravindra Mukund Malkar	International Advanced Research Journal in Science, Engineering and Technology, Vol. 7, Issue 12, December 2020	Various methodologies including Support Vector Machine (SVM), Fuzzy C-means Clustering, and genetic algorithms have been explored for electricity theft detection. Integration of smart meters and advanced algorithms offers potential solutions to minimize non-technical losses in distribution networks.	Accuracy and infrastructure requirements pose challenges in implementing existing theft detection systems. Techniques like SVM may suffer from accuracy issues, especially in classifying data accurately.
Title: Power Theft Detection System (Review Paper) Authors: Papai Bhakta, Suman Debnath, Partha Debnath, Paramita Das, Suparna Pal	International Journal of Creative Research Thoughts (IJCRT), Volume 10, Issue 5, May 2022	Electricity theft, in various forms such as meter tampering and illegal connections, poses significant challenges to power utilities. Traditional methods of detecting power theft are being replaced by modern technologies like microcontrollers and smart meters.	Detection of power theft using traditional methods relies heavily on manual operations, leading to inefficiencies and inaccuracies. The installation and maintenance costs associated with implementing smart meters and advanced detection systems may pose financial challenges for power utilities.
Title: Power Theft Detection using GSM Authors: Rekha R. Patil, Shwetha, Thrupthi, Vaishnavi S. Shetty, Sanjeevi Kumar P	International Journal of Engineering Research & Technology (IJERT) / 2019	The proposed system effectively detects and prevents electricity theft by using a combination of hardware components such as leaf switches, current transformers, microcontrollers, relays, LCD displays, and GSM modules. The system offers real-time monitoring and notification capabilities, allowing authorities to take necessary action upon detecting theft attempts.	The paper does not extensively discuss the scalability of the proposed system, particularly in larger utility networks or urban areas with high electricity consumption. The experimental setup and findings may not fully capture the real-world challenges and complexities associated with electricity theft detection in diverse environments.
Title: Review Paper on Power Theft Detection and Prevention Authors: Prof. Rupali Shinde, Prof. Soniya Joshi, Vaibhav Rampure, Rajendra Wagh, Dinesh Potdar, Mohsin Jamadar	International Journal of Advance Research, Ideas and Innovation in Technology (IJARIIE), Volume 6, Issue 5, 2020	Various techniques and systems have been proposed for the detection and prevention of power theft, utilizing components such as IoT, Arduino, GSM, PIC Microcontroller, current sensors, and voltage sensors. The reviewed systems aim to minimize the illegal use of electricity, reduce the chances of theft, and take appropriate actions in case of theft detection.	Some systems rely heavily on specific technologies like GSM or IoT, which may have limitations in terms of coverage, reliability, or compatibility with existing infrastructure. Practical testing and validation of these systems in diverse environments, including urban, rural, and industrial settings, are essential to assess their effectiveness and address potential issues.

EXISTING SYSTEM

In existing systems errors may occur due to less concentration. It requires Huge Man power .

- The system monitoring work gets delayed due to external conditions. Due to this Manual operation there is a high wastage of Power.

- Energy crisis is the main problem that our society faces. A relevant system control and monitor power usage is one of the solutions to this problem.

- One approach which is today's energy crisis that can be addressed through reduction of power usage in households.

PROPOSED SYSTEM

- ## Hardware Components

The system integrates two ESP8266 modules and two current sensors.

- ## Notification System

The System sends the notification when the theft is done.

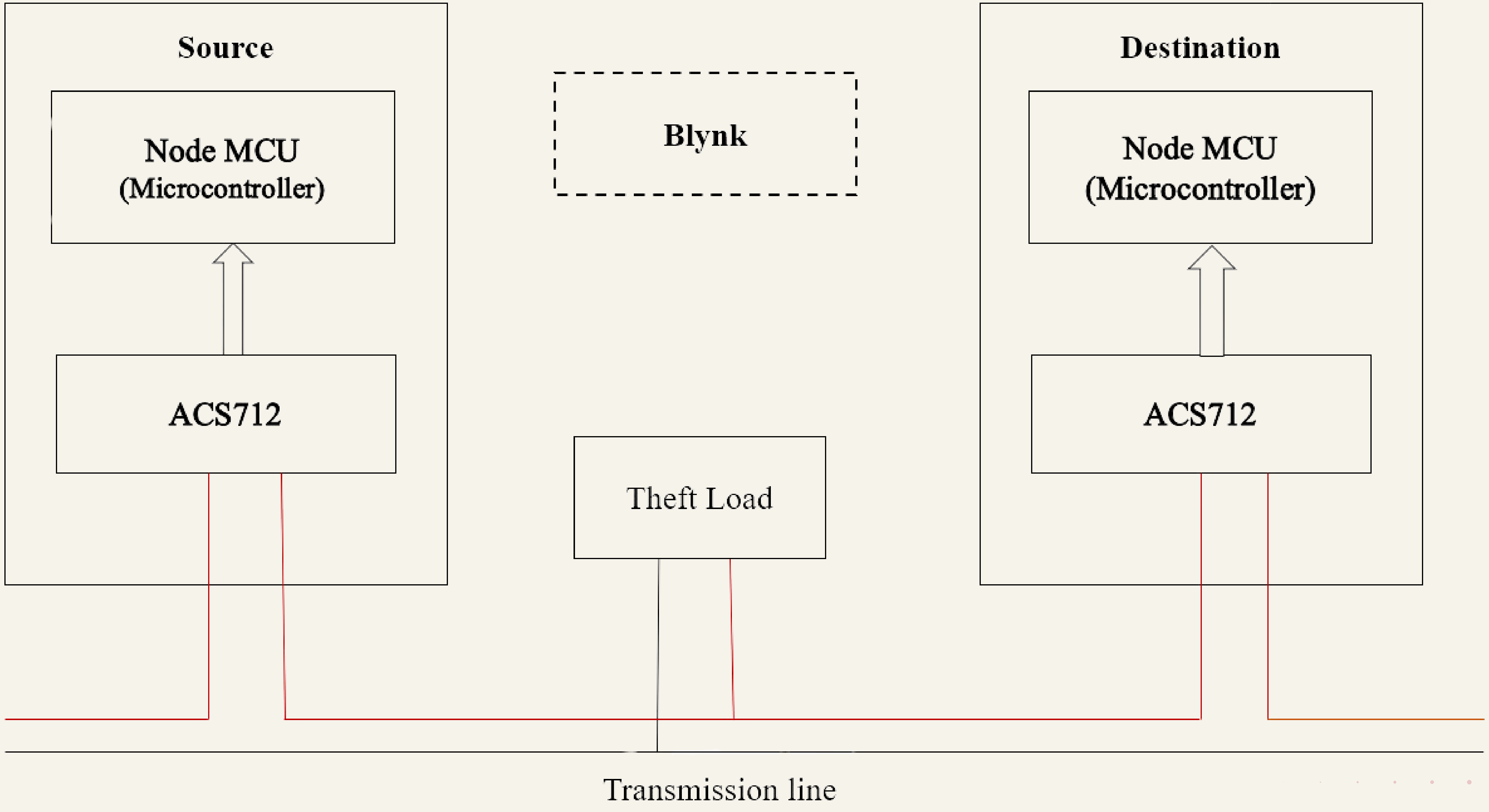
- ## Blynk IoT Platform

The system utilizes the Blynk app for remote monitoring and control.

- ## Theft Load

This enables to detect and respond to instances of unauthorized power consumption.

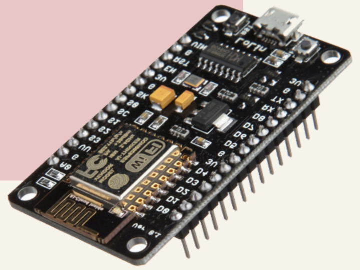
BLOCK DIAGRAM



COMPONENTS

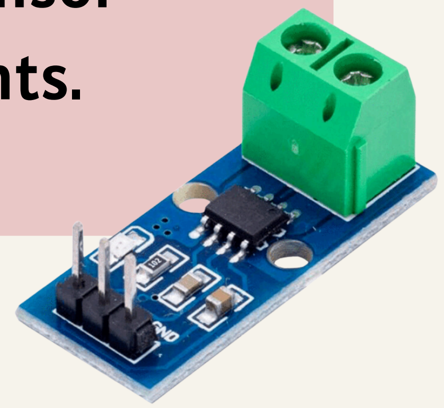
ESP8266

The ESP8266 is a Wi-Fi module that enables wireless communication and data transmission.



Current Sensors (ACS712)

The ACS712 is a Hall effect-based current sensor capable of measuring both AC and DC currents.



Blynk IoT Platform

Blynk is a popular IoT platform that allows users to control and monitor connected devices remotely through a mobile app.

Theft Load

This load is deliberately introduced to mimic scenarios where unauthorized devices or connections draw power from the distribution system.

FEATURES

The system features:

- Real-time monitoring of power consumption.
- Anomaly detection for potential theft.
- Integration with the Blynk IoT platform for remote access.
- Alert notifications for prompt action.
- Scalability for deployment across diverse networks.

APPLICATION

- Utility companies: Combat electricity theft, reduce revenue losses.
- Government agencies: Enforce energy regulations, prevent illegal activities.
- Smart cities: Ensure efficient, secure electricity distribution.
- Industrial facilities: Optimize energy usage, minimize costs.
- Residential/commercial buildings: Monitor electricity usage, detect inefficiencies.
- Renewable energy projects: Ensure integrity, reliability of energy sources.

PARAMETERS

HARDWARE DESCRIPTION

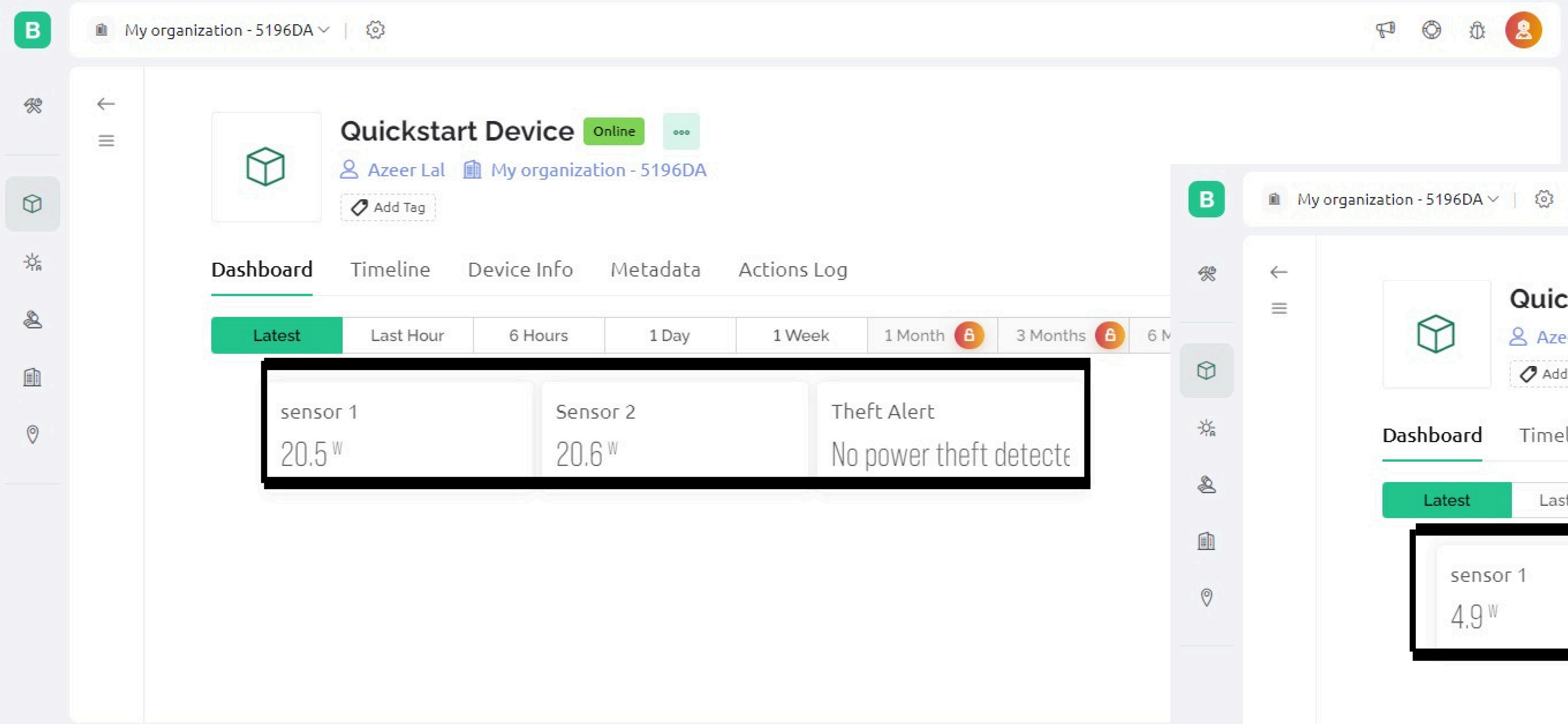
- ESP8266
- Current Sensor(ACS712)
- Additional Load for detection

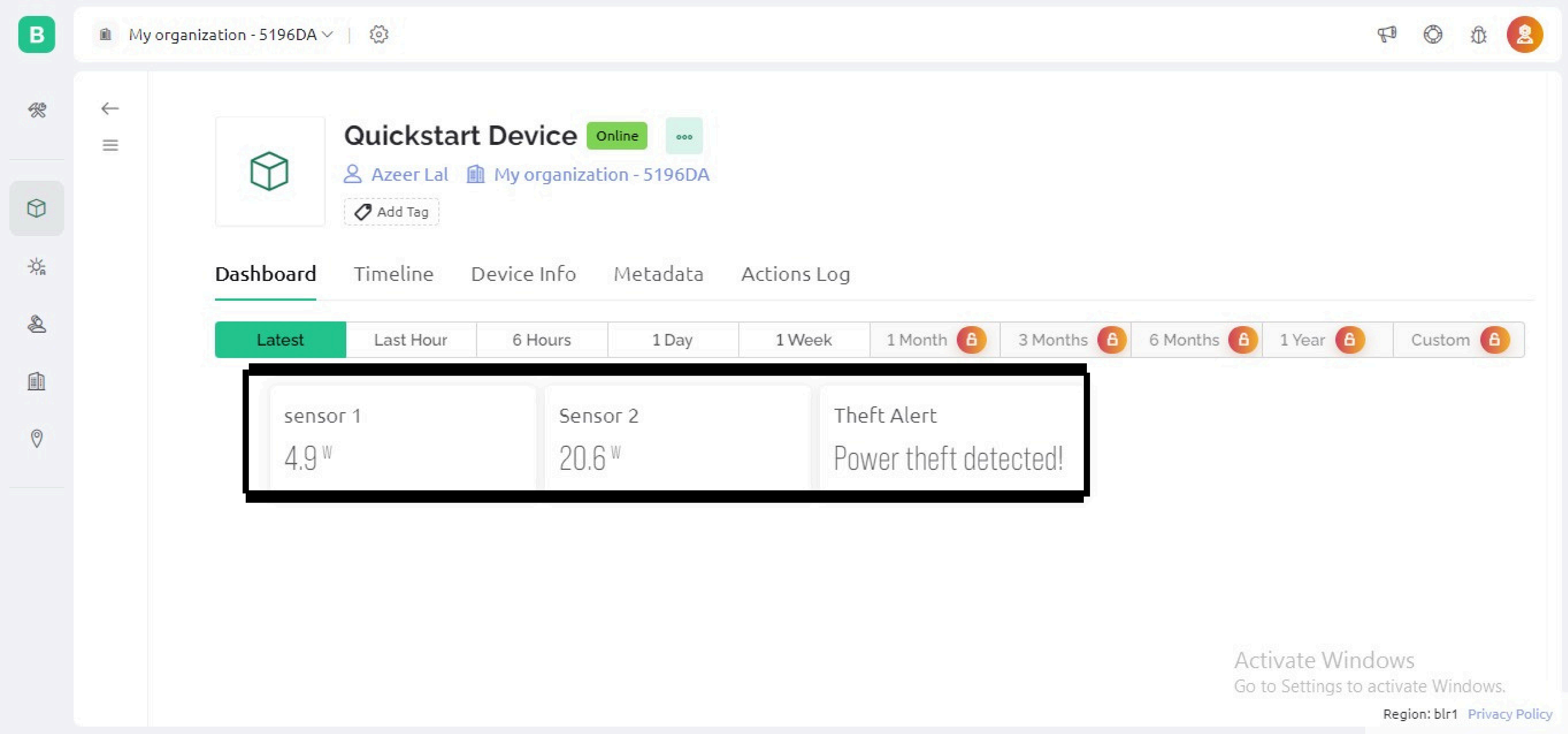
SOFTWARE DESCRIPTION

- Arduino IDE
- Blynk application
- Pushbullet

OUTPUT AND RESULT

The system accurately identifies irregularities in power consumption patterns, such as sudden spikes or drops, signaling potential electricity theft.





CONCLUSION

This initiative is geared towards boosting IoT connectivity and networking within the realm of smart city evolution. We're deploying sensors to detect facility theft, pinpoint faults, and track the fault area for necessary actions. The proposed system addresses key challenges in India's current grid setup, such as energy wastage, power theft, and cable faults. Moreover, it helps curb factors prompting electricity theft, contributing to a more robust and accountable energy framework.

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